

Large-Scale Object Detection in the Wild With Imbalanced Data Distribution, and Multi-Labels

Cong Pan , Junran Peng , Xingyuan Bu, and Zhaoxiang Zhang , *Senior Member, IEEE*

Abstract—Training with more data has always been the most stable and effective way of improving performance in the deep learning era. The Open Images dataset, the largest object detection dataset, presents significant opportunities and challenges for general and sophisticated scenarios. However, its semi-automatic collection and labeling process, designed to manage the huge data scale, leads to label-related problems, including explicit or implicit multiple labels per object and highly imbalanced label distribution. In this work, we quantitatively analyze the major problems in large-scale object detection and provide a detailed yet comprehensive demonstration of our solutions. First, we design a concurrent softmax to handle the multi-label problems in object detection and propose a soft-balance sampling method with a hybrid training scheduler to address the label imbalance. This approach yields a notable improvement of 3.34 points, achieving the best single-model performance with a mAP of 60.90% on the public object detection test set of Open Images. Then, we introduce a well-designed ensemble mechanism that substantially enhances the performance of the single model, achieving an overall mAP of 67.17%, which is 4.29 points higher than the best result from the Open Images public test 2018.

Index Terms—Computer vision, object detection, long-tail distribution, noisy labels, multi-labels.

I. INTRODUCTION

DATA is playing a primary and decisive role in deep learning. With the advent of the ImageNet dataset [1], deep

Manuscript received 14 July 2023; revised 4 June 2024; accepted 15 June 2024. Date of publication 1 July 2024; date of current version 5 November 2024. This work was supported in part by the National Key R&D Program of China under Grant 2022ZD0116500, in part by the National Natural Science Foundation of China under Grant U21B2042 and Grant 62320106010, and in part by the 2035 Innovation Program of CAS, and the InnoHK program. Recommended for acceptance by A. Kovashka. (Cong Pan and Junran Peng are Co-first authors.) (Corresponding author: Zhaoxiang Zhang.)

Cong Pan is with the New Laboratory of Pattern Recognition (NLPR), State Key Laboratory of Multimodal Artificial Intelligence Systems (MAIS), Institute of Automation, Chinese Academy of Sciences (CASIA), Beijing 100190, China, and also with the School of Future Technology, University of Chinese Academy of Sciences (UCAS), Beijing 100049, China (e-mail: pancongpc@163.com).

Junran Peng is with the Institute of Automation, Chinese Academy of Sciences (CASIA), Beijing 100190, China (e-mail: jrpeng4ever@126.com).

Xingyuan Bu is with Alibaba Group, Hangzhou 311121, China (e-mail: buxingyuan.bxy@taobao.com).

Zhaoxiang Zhang is with the New Laboratory of Pattern Recognition (NLPR), State Key Laboratory of Multimodal Artificial Intelligence Systems (MAIS), Institute of Automation, Chinese Academy of Sciences (CASIA), Beijing 100190, China, also with the University of Chinese Academy of Sciences (UCAS), Beijing 100049, China, and also with the Centre for Artificial Intelligence and Robotics Hong Kong Institute of Science and Innovation, Chinese Academy of Sciences (HKISI-CAS), Hong Kong 999077, China (e-mail: zhaoxiang.zhang@ia.ac.cn).

Our result is published on <https://www.kaggle.com/c/open-images-2019-object-detection/leaderboard>.

Digital Object Identifier 10.1109/TPAMI.2024.3421300

neural network [2] was well-exploited for the first time, leading to an unprecedented surge in deep learning research. Some recent works [3], [4] also prove that larger quantities of data with lower-quality labels (such as hashtags) can significantly surpass state-of-the-art methods by a large margin. Throughout the history of deep learning, it is evident that the development of the field is closely tied to the availability and quality of data.

In the past years, significant progress has been made in the field of object detection. Some generic object detection datasets with annotations of high quality like Pascal VOC [5] and MS COCO [6] greatly boost the development of object detection, giving birth to plenty of amazing methods [7], [8], [9], [10], [11], [12], [13]. However, these datasets are considered relatively small by today's standards and are beginning to limit the advancement of object detection area to some extent. Consequently, attempts are frequently made to focus on atomic problems on these datasets instead of exploring object detection in more challenging scenarios.

Recently, the Open Images V4 [14] dataset has been released, comprising 1.7 million images with 12.4 million annotated bounding boxes across 500 categories. This unseals the limits of data-hungry methods and may stimulate research to bring object detection to more general and sophisticated situations. However, accurately annotating data on such a scale is labor-intensive, making manual labeling almost infeasible. The annotating procedure of the Open Images dataset is completed with the strong assistance of deep learning that candidate labels are generated by models and verified by humans. This approach inevitably weakens the quality of labels because of the uncertainty of models and the knowledge limitation of human individuals, which leads to several major problems.

Objects in the Open Image dataset may explicitly or implicitly have multiple labels, which differs from traditional object detection. The object classes in Open Images form a hierarchy that most objects potentially hold a leaf label and all corresponding parent labels. However, due to the annotation quality, there are cases where objects are only labeled as parent classes, and leaf classes are absent. Apart from hierarchical labels, objects in the Open Images dataset may also hold several leaf classes, such as *car* and *toy*. Another annoying case is that objects of similar classes are frequently annotated as each other in both training and validation sets, for example, *torch* and *flashlight*, as shown in Fig. 1.

As images in the Open Images dataset are collected from open source in the wild, the label distribution is extremely imbalanced, including both very frequent and infrequent classes.

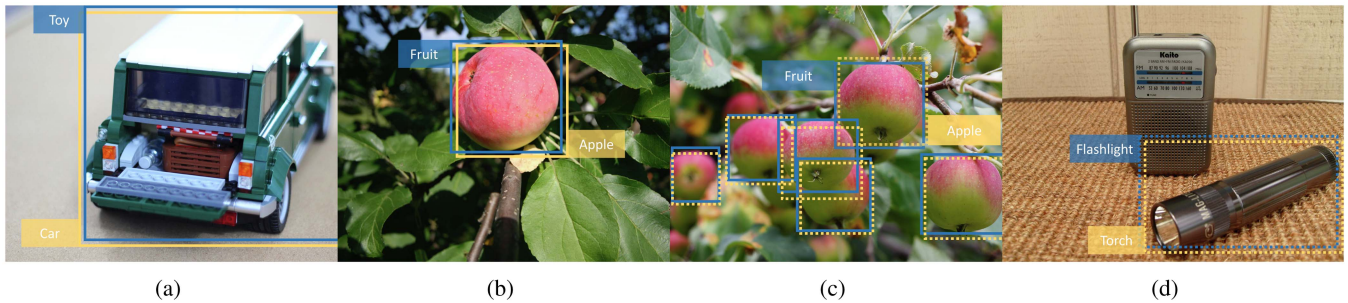


Fig. 1. Examples of multi-label annotations in the Open Images dataset. (a) and (b) are cases where objects are explicitly annotated with multiple labels. In (a), a toy car is labeled as both *car* and *toy*. In (b), an apple is hierarchically labeled as *apple* and *fruit*. (c) and (d) are cases where objects implicitly have multiple labels. In (c), apples are only labeled as *fruit*. In (d), flashlights are always randomly labeled as either *torch* or *flashlight*.

Consequently, methods for balancing label distribution are required to train detectors more effectively. Nevertheless, earlier methods such as over-sampling tend to impose over-fitting on infrequent categories and fail to fully use the data of frequent categories.

In this work, we engage in solving these major problems in large-scale object detection. Our contributions are threefold: (1) We design a concurrent softmax to deal with the explicitly and implicitly multi-label problem. Different from our first version reported in the original conference publication [15], we provide more qualitative visualization results and reveal mAP gains for single classes affected by the co-occurrence phenomenon. (2) We propose a soft-balance method together with a hybrid training scheduler to mitigate over-fitting on infrequent categories and better exploit data of frequent categories. We further merge the two training schemes together as one and this unification brings great robustness, which is new compared to our conference version [15]. (3) We provide detailed ablation studies on our final ensemble strategy, which are abundant new contents. The concurrent softmax and sampling method yield an overall gain of 3.34 points, leading to an mAP of 60.90% for a single model result on the public test-challenge set of Open Images. This is 5.09 points higher than the single model result of the first-place method [16] on the public test-challenge set from the previous year. More importantly, our ensemble strategy and additional techniques further boost our results by 6.27 points, achieving an overall mAP of 67.17% on the public leaderboard of the Open Image Detection Challenge 2019.

II. RELATED WORKS

A. Object Detection

Generic object detection is a fundamental and challenging problem in computer vision, with numerous works [7], [8], [9], [10], [11], [13], [17], [18], [19], [20], [21], [21], [22], [23], [24], [25] emerging in recent years. Faster-RCNN [7] first proposes an end-to-end two-stage framework for object detection and lays a strong foundation for successive methods. YOLO [8] and SSD [9] are one-stage detectors designed for real-time and easily-deployed object detection. Previous methods primarily use anchor boxes to provide classification and regression references. Some anchor-free approaches, such as CornerNet [17]

and CenterNet [18], directly predict object locations, sizes, and classes to simplify the detection pipeline and potentially reduce the complexity and computational overhead associated with anchor-based methods. [19], [20] focus on predicting higher-quality boxes to accommodate the COCO metric.

In addition, several methods are dedicated to solving the scale problem in object detection. In [21], deformable convolution is proposed to adaptively sample input features to deal with objects of various scales and shapes. [22], [23], [26] utilize dilated convolutions to enlarge the effective receptive fields of detectors to better recognize objects of different scales. Transformer-based detectors, such as DETR [11] and Deformable DETR [12], have also gained popularity by leveraging the self-attention mechanism of transformers to model long-range dependencies, achieving impressive results, especially in complex scenes with multiple objects. Furthermore, post-processing-free methods remove Non-Maximum Suppression (NMS) to achieve a more streamlined and efficient detection pipeline. For example, YOLOX [24] is a prominent NMS-free object detector that integrates the elimination of duplicate detections directly into the network, simplifying the inference process and improving overall performance.

However, most works are still exploring datasets like Pascal VOC and MS COCO, which are small by modern standards. Only a few works [16], [27], [28], [29] have been proposed to deal with large-scale object detection datasets like the Open Images dataset. Wu et al. [28] propose a soft box-sampling method to cope with the partial label problem in large-scale datasets. In [27], a part-aware sampling method is designed to capture the relationships between entities and parts to aid in the recognition of part objects.

While the methods mentioned above strive to solve the problems of sparse annotation and class imbalance in the Open Images dataset, they often overlook crucial aspects such as the presence of implicit or explicit multi-labels within the dataset. In contrast, our work delves into the intricacies of the Open Images dataset by addressing the issues of implicit and explicit multi-labels, as well as the imbalance in label distribution. Recognizing that these factors play a significant role in the overall detection accuracy, we propose targeted solutions to tackle these key pain points. By implementing these solutions, we have effectively enhanced the detection accuracy on

large-scale object detection datasets, demonstrating the importance of addressing these often-neglected aspects in improving overall model performance.

B. Multi-Label Recognition

There have been many notable attempts [30], [31], [32], [33], [34], [35], [36], [37], [38] to solve the multi-label classification problem from various perspectives. One simple and intuitive approach [39] is to transform the multi-label classification problem into multiple binary classification problems and fuse the results, but this method neglects relationships between labels. Some works [34], [40], [41], [42] embed dependencies among labels with deep learning to improve the performance of multi-label recognition. In [33], [34], [35], [36], [43], graph structures are utilized to model the label dependencies. Gong et al. [30] use a ranking-based learning strategy and reweight losses of different labels to achieve better accuracy. Moreover, methods in [31], [37], [44] propose RNN or CNN-RNN hybrid frameworks to embed labels into latent space to capture the correlation between them. Furthermore, [45], [46], and [38] leverage Transformers to exploit the complex dependencies among visual features and labels.

However, our proposed method is specifically tailored to address the multi-label problem in large-scale object detection tasks, distinguishing it from existing approaches in multi-label recognition. Unlike image classification, where the ranking of scores for different classes in each image is crucial, object detection relies heavily on mean average precision, where the absolute score values of predicted bounding boxes play a pivotal role in determining their ranking across the entire dataset. This fundamental difference underscores the necessity of our proposed approach.

C. Imbalanced Label Distribution

There have been numerous efforts to address long-tailed label distribution. These methods can be divided into three types: data-based re-sampling strategies [3], [29], [47], [48], [49], [50], loss-based methods [51], [52], [53], [54], [55], [56], and data augmentation methods [57], [58], [59], [60], [61]. In [29], [47], [49], class-aware sampling is applied, ensuring that each mini-batch is filled as uniformly as possible with respect to different classes. [48], [50] expand samples of minor classes through synthesizing new data. Dhruv et al. [3] compute a replication factor for each image based on the distribution of hashtags and duplicate images the prescribed number of times. BBN [62] unifies the representation learning on non-balanced data samplers and re-balanced data samplers, which effectively avoids overfitting on infrequent categories. As for loss-based methods, loss weights are assigned to samples of different classes to match the imbalanced label distribution. In [52], [53], [54], samples are reweighted by inverse class frequency, while Yin et al. [51] calculate the effective number of each class to re-balance the loss. In OHEM [63] and focal loss [10], the difficulty of samples is evaluated in terms of losses, and hard samples are assigned

higher loss weights. In [64], they design a label-distribution-aware loss to enlarge the margins for minority classes. EFL [65] introduces a class-wise equalization mechanism that dynamically adjusts the loss weights for each class, aiming to mitigate the class imbalance issue and promote more balanced learning across different object categories. As for data augmentation methods, [60] explores the feasibility of data mixup technology in long-tailed recognition algorithms and finds that data mixup can prevent overfitting of the network and enhance its generalization ability. [61] addresses the data scarcity issue of the rare class by feature augmentation and sampling adaption. [66], [67] and [68] provide more thorough summaries of related works.

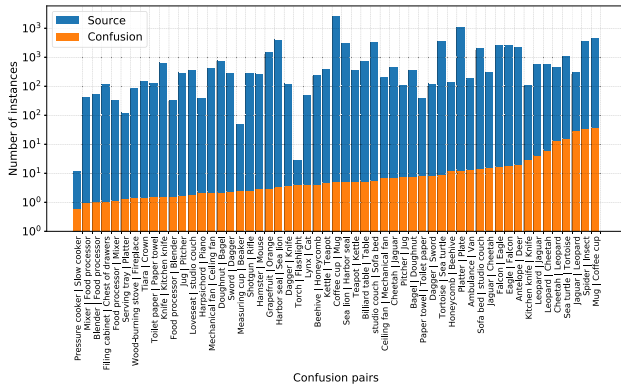
However, the re-sampling methods mentioned above can lead to the under-sampling of frequent categories and the over-sampling of infrequent categories. To solve this problem, we propose a soft-balance method with a hybrid training scheduler that simultaneously ensures sufficient training of dominant categories while reducing the over-sampling of infrequent categories.

III. PROBLEM SETTING

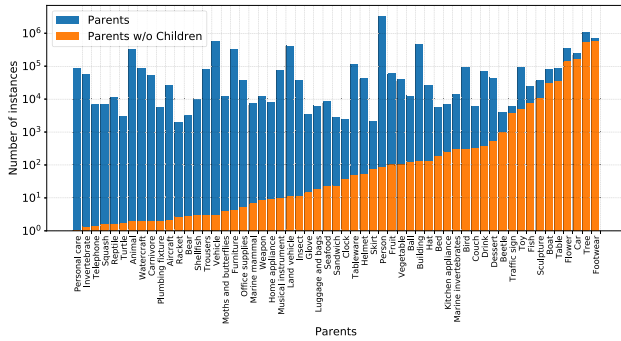
The Open Images dataset is currently the largest released object detection dataset to the best of our knowledge. It contains 12 million annotated instances across 500 categories on 1.7 million images. Owing to its scale size and annotation styles, we argue that there are three major problems regarding this kind of dataset, apart from the missing annotation problem which has been discussed in [27], [28].

Objects may explicitly have multiple labels: As objects in the physical world often possess rich categorical attributes on different levels, the 500 categories in the Open Images dataset form a class hierarchy of 5 levels, with 443 of the nodes as leaf classes and 57 as parent classes. It is common for some objects to hold multiple labels, including both leaf labels and parent labels, such as *apple* and *fruit* as shown in Fig. 1(b). Another case is that an object could easily have multiple leaf labels. For instance, a car-toy is labeled as *toy* and *car* at the same time as shown in Fig. 1(a). This happens frequently in the dataset and all leaf labels are requested to be predicted during evaluation. Different from previous single-label object detection, how to deal with multiple labels is one of the crucial factors for object detection in this dataset.

Objects may implicitly have multiple labels: Beyond the explicit multi-label problem, there is also an implicit multi-label problem caused by the limited and inconsistent knowledge of human annotators. Many pairs of leaf categories are challenging to distinguish, leading to randomly mixed labels. We analyze the proportion that an object of a leaf class is labeled as another and find that there are at least 115 pairs of severely confused categories (confusion ratio ≥ 0.1). We display the top-55 confused pairs in Fig. 2(a), and find many categories heavily confused. For instance, nearly 65% of the torches are labeled as *flashlight*, and 50% of the leopards are labeled as *cheetah*.



(a) We select the top-55 confused category pairs and show their concurrent rates.



(b) We show the ratio of parent annotations without leaf labels and total parent annotations.

Fig. 2. Implicit multi-label problem caused by confused categories and absence of leaf classes.

Besides, labels of leaf and parent classes are often incomplete, resulting in a large number of objects being annotated only with parent labels, lacking specific leaf labels. As shown in Fig. 1(c), an apple is sometimes labeled as *apple* and at other times labeled as *fruit* without the leaf annotation. We demonstrate the ratio of leaf annotations and parent annotations lacking leaf annotations in Fig. 2(b). This implicit co-existence phenomenon also happens frequently and needs to be taken into consideration, otherwise detectors may learn incorrect signals.

Imbalanced label distribution: To build such a huge dataset, images are collected from open resources in the wild. As one can expect, the Open Images dataset suffers from an extremely imbalanced label distribution, both infrequent and very frequent categories are included. As shown in Fig. 3, the most frequent category owns nearly 30 k times the number of training images as the least frequent category. A naive re-balance strategy, such as the widely used class-aware sampling [29], which uniformly samples training images from different categories, cannot effectively address such extreme imbalance. This approach may lead to two significant consequences:

- 1) For frequent categories, they are not trained sufficiently, for the reason that most of their training samples have never been seen and are wasted.
- 2) For infrequent categories, excessive over-sampling may cause severe over-fitting and degrade the generalizability of recognition on these classes. There are two reasons for this:

- (1) Oversampling of infrequent classes can potentially increase the model's variance because it introduces redundant or synthetic samples that might not accurately represent the true underlying distribution. This increased variance can contribute to overfitting.
- (2) Oversampling of infrequent classes can affect the decision boundaries of a model, making it more sensitive to variations in the minority class. In extreme cases, the model might focus too much on these rare instances, disregarding the majority classes, which can lead to poor generalization.

Once adopting the class-aware sampling, a category like *person* is extremely undersampled that 99.13% of the instances are neglected, while a category like *pressure cooker* is immensely oversampled that each instance is seen for 252 more times averagely within an epoch.

IV. METHODOLOGY

In this section, we explore methods to deal with label-related problems in large-scale object detection. First, we design a *concurrent softmax* to handle both explicit and implicit multi-label issues jointly. Second, we propose a *soft-balance sampling* combined with a *hybrid training scheduler* to deal with the extremely imbalanced label distribution.

A. Multi-Label Object Detection

As one of the most widely used loss functions in deep learning, the form of softmax loss for a bounding box b is presented as follows:

$$L_{cls}(b) = - \sum_{i=1}^C y_i \log(\sigma_i), \quad \text{with } \sigma_i = \frac{e^{z_i}}{\sum_{j=1}^C e^{z_j}}, \quad (1)$$

where z_i denotes the response of the i -th class, y_i denotes the label, and C means the number of categories. It behaves well in single-label recognition where $\sum y_i = 1$. It enables the response of the ground-truth class to suppress others and is friendly to optimization, which are reasons why softmax with cross-entropy is more popular than sigmoid with binary cross-entropy in single-label recognition.

However, things are different when it comes to multi-label recognition. In the conventional object detection training scheme, each bounding box is assigned only one label during training, ignoring other ground-truth labels. If we force to assign all $m(m \geq 1)$ ground-truth labels belonging to $K = \{k \mid k \in C, y_k = 1\}$ to a bounding box during training, the scores of multiple labels would restrain each other. When computing the gradient of each ground-truth label, it looks like the below:

$$\frac{\partial L_{cls}}{\partial z_i} = \begin{cases} m\sigma_i - 1, & \text{if } i \in K; \\ m\sigma_i, & \text{if } i \notin K. \end{cases} \quad (2)$$

When $m\sigma_i > 1$ for $i \in K$, z_i is optimized to become lower even if i is one of the ground-truth labels, which is the wrong optimization direction.

The influence is even more prominent during testing than training in the task of object detection. In image classification, the main evaluation metric is precision which counts the ratio of true positives over all results. Therefore, the ranks of scores of different classes within each single image are more important

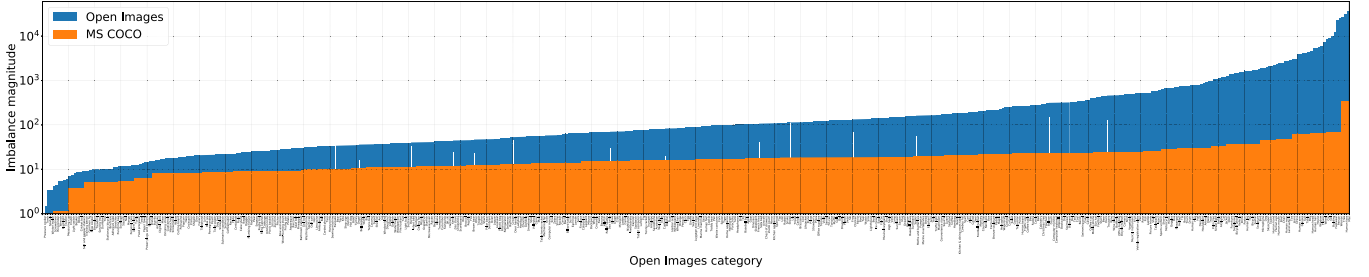


Fig. 3. The imbalance magnitude of the Open Images and the MS COCO dataset. *Imbalance magnitude* means the number of images of the largest category divided by the smallest. (Best viewed on high-resolution display.)

rather than the absolute value of the score itself. However, in object detection, the most common evaluation metric is mean average precision. For each class, the detection results of all images are collected and ranked together, and the area of the precision-recall curve is calculated as the average precision. In this way, the absolute value of the score per box matters, because it may influence the rank of a predicted box result over the entire dataset.

1) *Concurrent Softmax*: The *concurrent softmax* is designed to help solve the problem of recognizing objects with multiple labels in object detection. During training, the concurrent softmax loss of a predicted box b is presented as follows:

$$L_{cls}^*(b) = - \sum_{i=1}^C y_i \log \sigma_i^*,$$

$$\text{with } \sigma_i^* = \frac{e^{z_i}}{\sum_{j=1}^C (1 - y_j)(1 - r_{ij})e^{z_j} + e^{z_i}}, \quad (3)$$

where y_i denotes the label of class i regarding the box b , and r_{ij} denotes the concurrent rate of class i to class j . The output of concurrent softmax during training is defined as:

$$\frac{\partial L_{cls}^*}{\partial z_i} = \begin{cases} \sigma_i^* - 1, & \text{if } i \in K; \\ \sum_{j \in K} (1 - r_{ij}) \sigma_i^*, & \text{if } i \notin K. \end{cases} \quad (4)$$

Unlike softmax, where the responses of the ground-truth categories suppress all others, we remove the suppression effects between explicitly coexisting categories in concurrent softmax. For instance, a bounding box is assigned multiple ground-truth labels $K = \{k \mid k \in C, y_k = 1\}$ during training. When computing the score of class $i \in K$, the influences of all the other ground-truth classes $j \in K \setminus \{i\}$ are neglected because of the $(1 - y_j)$ term, and the score of each correct class is boosted. This approach avoids the unnecessarily large losses caused by the multi-label problem, allowing the gradients to focus on more valuable knowledge.

Apart from explicit coexistence cases, there remain implicit concurrent relationships to be addressed. As mentioned in Section III, there are two types of implicit concurrent relationships: hierarchical relationships and confusion relationships. For hierarchical relationships, objects are sometimes only annotated with a parent label without a leaf label, such as an apple being labeled as *fruit* without the leaf *apple* annotation. The other case is confusion relationships, where objects of similar classes are

labeled as each other with a certain probability, such as *torch* and *flashlight*.

We define a concurrent rate r_{ij} as the probability that an object of class i is labeled as class j . In our setting, two labels are recognized to be concurrent as long as the IoU of the two bounding boxes is greater than a concurrent threshold τ , which is a certain value. The value of r_{ij} is calculated based on the class annotations of the training set as follows:

$$r_{ij} = \frac{\text{len}(\text{IoU}(B_i, B_j) > \tau)}{\text{len}(B_i)}, \quad (5)$$

where $\text{len}(\cdot)$ denotes the number of elements. The choice of the fixed concurrent threshold τ depends on the detector. For example, in this paper, we employ the Faster R-CNN two-stage detector, where the NMS threshold θ in the Region Proposal Network (RPN) is set to 0.7. Consequently, τ should be chosen to be greater than the NMS threshold and less than 1. We set $\tau = 0.9$ (similar final results are achieved in the range of $0.85 \sim 0.9$) in our experiments for the following reasons:

- 1) If τ is set too high (close to 1), it may lead to the exclusion of valid concurrent labels. This may lead to an underestimation of the true concurrency rate and fail to solve the multi-label suppression problem.
- 2) If τ is set too low (close to the NMS threshold), it may include pairs of bounding boxes that are not truly concurrent, leading to an overestimation of the concurrency rate. This can cause the model to incorrectly learn to associate unrelated object classes, thereby reducing the overall detection accuracy.

Therefore, the choice of $\tau = \frac{\theta+1}{2}$ is a balance between these two considerations, aiming to accurately capture the concurrency relationship between object classes while minimizing the impact on detection performance. The concurrent rate can capture the likelihood of co-occurrence and spatial overlap between different object classes in the dataset. By quantifying this relationship, the model can better understand and predict the presence of multiple labels for a single object, which is particularly important in scenarios where objects are closely located or partially occluded. This approach helps in addressing the multi-label suppression issue commonly encountered in object detection tasks.

Fig. 2(a) shows the concurrent rates of confusion pairs. Note that r_{ij} is set to 0 when class- i and class- j have hierarchical relationships, which means neither a child category should suppress

TABLE I
THE COMPARISON OF DIFFERENT LOSS FUNCTION METHODS

Loss Type	Balance	mAP
Focal Loss [10]	✓	50.18
BCE Loss	✓	54.29
Co-BCE Loss [16]	✓	55.74
dist-CE Loss [3]	✓	55.90
Softmax Loss		38.16
Concurrent Softmax Loss		39.14
Softmax Loss	✓	55.45
Concurrent Softmax Loss	✓	56.58

Models are trained in *mini-train* and evaluated on *mini-val*.

its parent nor a parent should suppress all its children. With the $(1 - r_{ij})$ term, suppression effects between confusing pairs are weakened.

The influence of multi-label object detection is also prominent during inference. Different from conventional multi-label recognition tasks, the evaluation metric of object detection is mean average precision (mAP). For each category, detection results for all images are first collected and ranked by scores to form a precision-recall curve, and the average area under the precision-recall curve is defined as the mAP. In this way, the absolute value of box score matters, as it may influence the rank of the predicted box across the entire dataset. Thus we also apply the concurrent softmax during inference, which is presented as follows:

$$\sigma_i^\dagger = \frac{e^{z_i}}{\sum_{j=1}^C (1 - r_{ij})e^{z_j} + e^{z_i}}, \quad (6)$$

where we abandon the $(1 - y_j)$ term and keep the concurrent rate term. The value of r_{ij} is the same probability obtained through statistical analysis of the training dataset. Scores of categories in a hierarchy and scores of similar categories would not suppress each other but are boosted effectively, which is desirable in the object detection task.

2) *Compared With BCE Loss*: Binary cross-entropy (BCE) is always a popular solution for multi-label recognition, but it does not work well in the multi-label detection task on the OpenImage dataset. We propose two possible reasons for this. First, both the sigmoid and focal loss are sensitive to the setting of hyper-parameters, especially in cases with long-tail data distribution. Second, the sigmoid function fails to normalize scores and declines the suppression effect between categories, which is desired when evaluated with the mAP metric. We have experimented with BCE loss and focal loss, but it turns out that they yield much worse results even than the original softmax loss, as shown in Table I.

3) *Retrospect the Issue of Noisy Multi-Labels*: As aforementioned in Section III, the noisy multi-labels issue in this paper refers to cases where objects are labeled as several confused categories or objects are labeled as child and parent categories at the same time. We take a dummy example to clarify the problem and how concurrent-softmax operates. As shown in Fig. 4, "a man in a monkey suit" may be annotated with the labels *woman*, *man*, *monkey*, and *person*, where *woman* and *man* are leaf labels under the *person* category. This instance is explicitly labeled as *man* and *person* concurrently. Utilizing

vanilla softmax may cause the activation of concurrent classes to compete with each other, resulting in diminished normalized values, such as the value of *person*. During the training stage with multiple positive labels, the activation values of correct predictions are even punished by supervision because of the interference of concurrent classes. With concurrent softmax, the interference between positive labels is removed while keeping suppressing the rest of the categories. In a noisy case where children labels are absent, which is prevalent in Open Images, the activation of *man* is inappropriately punished by vanilla softmax loss, while the concurrent softmax loss effectively prevents it.

B. Soft-Balance Sampling With Hybrid Training

As detailedly illuminated in Section III, the Open Images dataset suffers from a severely imbalanced label distribution. We denote by C the number of categories, N the total number of training images, and n_i the number of images containing objects of the i -th class. Conventionally, images are sampled in sequence without replacement for training in each epoch, and the original probability P_o of class i being sampled is denoted as $P_o(i) = \frac{n_i}{N}$, which may greatly degrade the recognition capability of the model for infrequent classes. A widely used technique, *i.e.*, class-aware sampling mentioned in [29], [47], [49] is a naive solution to handle the class imbalance problem, in which categories are sampled uniformly in each batch. The class-aware sampling probability P_a of class i becomes $P_a(i) = \frac{1}{C}$. Yet this may cause heavy over-fitting on infrequent categories and insufficient training on frequent categories as aforementioned.

To alleviate the problems above, we first adjust the sampling probability based on the number of samples, which is called *soft-balance sampling*. We then define the $P_n(i) = \frac{n_i}{\sum_{i=1}^C n_j}$ as the approximation of non-balance sampling probability $P_o(i)$ for convenience. The sampling probability of class-aware balance can be reformulated as:

$$P_a(i) = \frac{1}{C} = \frac{1}{CP_n(i)} P_n(i) = \alpha P_n(i), \quad (7)$$

where the $\alpha = \frac{1}{CP_n(i)}$ can be regarded as a balance factor that is inversely proportional to the number of categories and the original sampling probability.

To reconcile the frequent and infrequent categories, we introduce soft-balance sampling by adjusting the balance factor with a new hyper-parameter λ :

$$\begin{aligned} P_s(i) &= \alpha^\lambda P_n(i) \\ &= P_a(i)^\lambda P_n(i)^{(1-\lambda)}. \end{aligned} \quad (8)$$

Note that $\lambda = 0$ corresponds to non-balance sampling and $\lambda = 1$ corresponds to class-aware balance. The normalized probability is:

$$P_s^*(i) = \frac{P_s(i)}{\sum_{j=1}^C P_s(j)}. \quad (9)$$

This sampling strategy guarantees more sufficient training on dominant categories and decreases the excessive sampling frequency of infrequent categories.

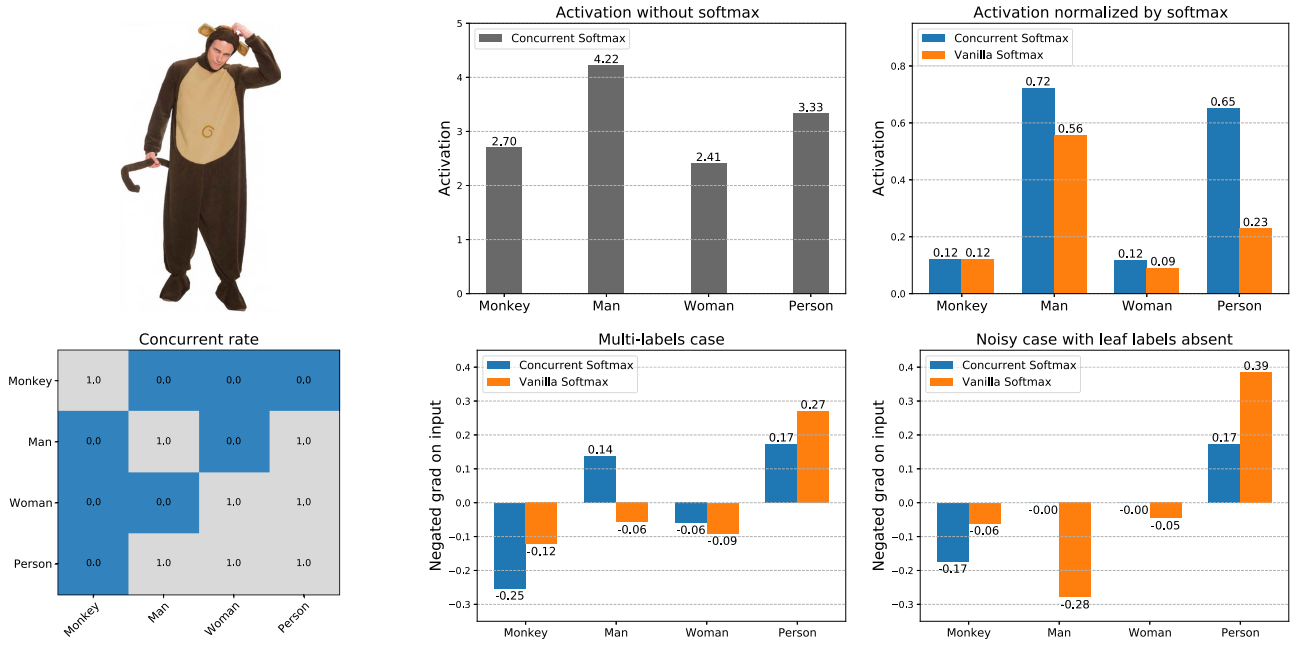


Fig. 4. An example demonstrating how concurrent softmax works. *Top-left*: A man in monkey suit, labeled as *man* and *person*. *Top-middle*: Raw activation values without softmax. *Top-right*: Activation values normalized by softmax. *Bottom-left*: Concurrent rates in concurrent softmax. *Bottom-middle*: Negated gradients on raw activation when multi-labels exist. *Bottom-right*: Negated gradients on raw activation when leaf labels are absent.

Even with the soft-balance method, there are still many samples of frequent categories that are not sampled. Thus we propose a hybrid training scheduler to further mitigate this problem. We first train the detector using the conventional strategy, which involves sampling training images in sequence without replacement, and the equivalent sampling probability is P_o . We then fine-tune the model with the soft-balance strategy to cover categories with very few samples. This hybrid training schema exploits the effectiveness of a pre-trained model for the object detection task from Open Images itself rather than ImageNet. It ensures that all images are seen during training and endows the model with better generalization ability.

We further update our algorithm to unify the soft-balance sampling and hybrid training in a single form. We adopt a cosine-annealing λ to adjust the class-specific sampling probability according to the current training iterations, which is called the *cosine-annealing hybrid training*:

$$\lambda = \mathbb{1}(T_{cur} - T_{start}) \left(1 - \cos \left(\frac{T_{cur} - T_{start}}{T_{max} - T_{start}} \pi \right) \right) \frac{\lambda_0}{2}, \quad (10)$$

where λ_0 is a hyper-parameter for controlling the maximum value of λ , T_{start} is the starting point of applying cosine annealing sampling, T_{cur} is the current epoch and T_{max} is the total number of training epochs. At the beginning of training, we set $\lambda = 0$ to learn the universal pattern, and then gradually shift our focus to the tail data. Cosine annealing gradually increases the value of λ using the cosine function. In this function, as the epoch progresses, the value of λ first accelerates, then rises slowly, and finally accelerates again. This approach enables our network to initially learn more comprehensively and then focus

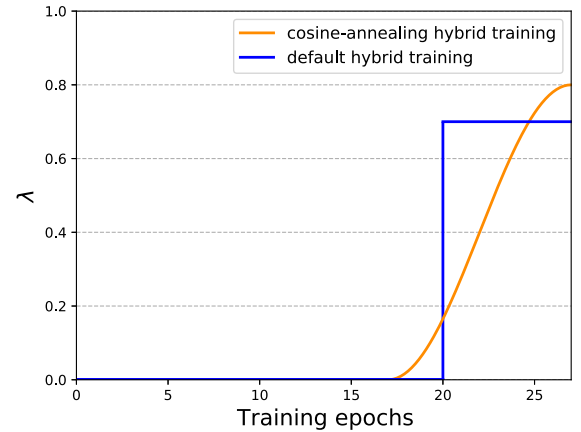


Fig. 5. The changing curve of λ in different methods.

on fine-tuning the learning, which can lead to better convergence and more accurate detection. As shown in Fig. 5, the model endures various sampling rates during training, which is more robust for hyper-parameter selection.

C. Model Ensemble

Model ensemble plays a prominent role in large-scale object detection like the Open Image Challenge. In this section, we aim to explore the potential benefits of Model Ensemble in further enhancing the performance of our proposed approach. While the primary focus of our paper is on addressing the challenges of class imbalance through Soft-balance Sampling and Concurrent

Algorithm 1: *box_merge*(b, B, w_b).

Require: b, B, w_b
 b is the target bounding box, in the form of (x_0, y_0, x_1, y_1) .
 B are set of bounding boxes to be merged, in the form of (x_0, y_0, x_1, y_1) .
 w_b denotes the merging weight.
Ensure: The merged box coordinates.
1: $b^* \leftarrow \text{mean}(B)$
2: **return** $(1 - w_b) \cdot b + w_b \cdot b^*$

Softmax, we consider Model Ensemble as an additional technique that can potentially complement and improve the results. Owing to the giant scale of data and numerous categories with complex hierarchical structures, even a giant model with plenty of parameters is far from handling well the problem. There are two steps of the model ensemble in our system. The first one is the ensemble on multi-scale results of a single model, while the other one is the ensemble on results of multi-models.

1) *Ensemble on Multi-Scale Results:* When merging the multi-scale results of each single model, we find that the conventional ensemble scheme works well which is concatenating the results of each test scale and then applying the vanilla NMS together with box voting. There are two crucial factors that have a profound influence on the results. Conventionally, only the top- n (usually 100 by default) boxes predicted by detectors are reserved, and boxes would be filtered by a score threshold t (0.1 by default). We find that increasing n and shrinking t both have a positive effect on the results. In our setting, we increase n from 100 to 1000 and increase t from 0.0 to 0.01 to avoid unlimited predicted boxes. Another interesting phenomenon is that the coherence of the training scale and testing scale matters, while different models may adapt to different scales. In our experiments, we keep the median scales during training and testing identical.

2) *Ensemble on Multi-Models:* Our ensemble strategy on multi-models involves three steps: (1) Re-score the test results of each model based on their per-class performance ranking. (2) We propose a special merging method that is highly effective. (3) We introduce a hierarchical merging scheme where model results are initially split into groups and merged into one, and then results from different groups are iteratively merged.

We first evaluate the per mAP result on the *val* dataset and collect the upper bound and lower bound across models. The re-score rate of each model is computed as follows:

$$r_i^c = \left(\frac{m_i^c - \alpha \min(m^c)}{\max(m^c) - m_i^c} \right)^\beta, \quad (11)$$

where m_i^c denotes the class- c mAP of model- i , α and β are hyper-parameters for adjusting the value. The output scores of class- c of each bounding box generated by model- i are rescaled by multiplying r_i^j .

When merging the results of different models, the algorithm we apply is as follows:

Algorithm 2: *score_merge*(s, S, w_s, n_s).

Require: b, B, w_s, n
 s is the score of the target bounding box.
 S are a set of scores of bounding boxes to be merged.
 w_s denotes the merging weight.
 n_s denotes the number of boxes to be selected.
Ensure: The merged box score.
1: $s^* \leftarrow \text{mean}(\text{topk}(S, n_s))$
2: **return** $s + w_s \cdot s^*$

Algorithm 3: Top-n Box Merging Algorithm.

Require: B, S, n_s, t, w_s, w_b
 B is a set of regression bounding boxes, in the form of (x_0, y_0, x_1, y_1) .
 S is a set of classification scores of bounding boxes.
 n_s is a hyper-parameter that represents the max number of boxes to be merged with.
 t_s is the NMS iou threshold for box filtering and score merging.
 t_b is the NMS iou threshold for box merging.
 w_s is a hyper-parameter that represents the adding weight of classification scores.
 w_b is a hyper-parameter that represents the merging weight for box regression.
 O denotes a list of the sorted index of boxes.
 K denotes a list of the kept index of boxes.
Ensure: Remained boxes coordinates and scores.
1: $K \leftarrow \{\}$;
2: $O \leftarrow \text{argsort } S$
3: **while** $O \neq \{\}$ **do**
4: $m \leftarrow O[0]$
5: $K \leftarrow K \cup m; O \leftarrow O - m$
6: $I_s \leftarrow \{i \in O | \text{iou}(b_m, b_i) \geq t_s\}$
7: $I_b \leftarrow \{i \in O | \text{iou}(b_m, b_i) \geq t_b\}$
8: $b_m \leftarrow \text{box_merge}(b_m, B[I_b], w_b)$
9: $s_m \leftarrow \text{score_merge}(s_m, S[I_s], w_s, n_s)$
10: $O \leftarrow O - I_s$
11: **end while**
12: **return** $B[K], S[K]$

Finally, we apply a l -level hierarchical merging strategy. In our setting, we find $l = 2$ works the best. We randomly split N models into $g = \text{round}(\sqrt{N})$ groups with each group roughly containing $\text{round}(N/g)$ models. We first ensemble the intra-group results and then ensemble the inter-group results.

V. EXPERIMENTS

A. Dataset

To analyze the proposed concurrent softmax loss and soft-balance with hybrid training, we conduct experiments on the Open Images Challenge 2019 dataset. As an object detection dataset in the wild, it contains 1.7 million images with 12.4 million boxes across 500 categories in its challenge split. The scale of training images is 15 times of the MS COCO [6] and 3 times

TABLE II
THE COMPARISON OF OPEN IMAGES AND OTHER OBJECT DETECTION DATASETS

Dataset	Pascal VOC	COCO	Obj365	Open Images
Classes	20	80	365	500
Images	11,540	123,287	638,606	1,784,662
Boxes	27,450	886,287	10,101,056	12,421,955
Boxes/Image	2.4	7.2	15.8	7.0
Classes/Image	1.5	2.9	5.0	2.2
Imba(image)	14.2	64.2	32,195.8	29,082.8
Imba(bbox)	15.9	136.0	75,746.3	101,328.1

Imba means the number of images or bounding boxes of the largest category divided by the smallest.

of the second-largest object detection dataset Object365 [69], as shown in Table II.

Considering the huge size of the Open Images dataset, we split a mini Open Images dataset for our ablation study. The mini Open Images dataset contains 115 K training images and 5 K validation images, named as *mini-train* and *mini-val*, respectively. All the images are sampled from the Open Images Challenge 2019 dataset with the ratio of each category unchanged. Final results on *full-val* and *public test-challenge* in the Open Images Challenge 2019 dataset are also reported. We follow the metric used in the Open Images challenge which is a variant mAP at IoU 0.5, as all false positives do not exist in image-level labels¹ are ignored.

B. Implementation Details

We train our detector with a ResNet-50 backbone armed with FPN for ablation studies on *mini-train*. For the network configuration, we follow the settings mentioned in Detectron [70]. We use SGD with a momentum of 0.9 and weight decay of 0.0001 to optimize parameters. The initial learning rate is set to 0.00125 $\times$ batch size and then decreased by a factor of 10 at 4 and 6 epochs for 1 $\times$ schedule, which has a total of 7 epochs. The input images are scaled so that the length of the shorter edge is randomly picked from [640, 672, 704, 736, 768, 800], and the longer side is limited to 1333. Horizontal flipping is used as data augmentation and sync-BN is adopted to speed up the convergence.

C. Concurrent Softmax

We explore the influence of concurrent softmax in the training and testing stages in this ablation study, respectively. All models are trained on *mini-train* and evaluated on *mini-val*.

The impacts of concurrent softmax during training: Table I shows the results of the proposed concurrent softmax compared with the vanilla softmax and other existing methods during the training stage. Concurrent softmax outperforms softmax by 1.13 points with class-aware sampling and 0.98 points with non-balance sampling. It is also found that sigmoid with BCE and focal loss behaves poorly in this case. We hypothesize that they are incompatible with the mAP metric in object detection

¹In the Open Images dataset, image-level labels consist of verified-exist labels and verified-not-exist labels. The unverified categories are ignored.

TABLE III
THE EFFECTIVENESS OF CONCURRENT SOFTMAX DURING TESTING

Train Method	Test Method	mAP
Softmax	Softmax	55.45
Softmax	Concurrent Softmax	55.77
Concurrent Softmax	Softmax	56.58
Concurrent Softmax	Concurrent Softmax	56.95

Models are trained on *mini-train* and evaluated on *mini-val*.

as mentioned in IV-A2. Our method also outperforms dist-CE loss [3] and Co-BCE loss [16].

The impacts of concurrent softmax during testing: We also show results to demonstrate the effectiveness of concurrent softmax in the testing stage in Table III. Solely applying concurrent softmax brings 0.36 mAP improvement during inference, while applying it in both the training and testing stages yields a total improvement of 1.50 points. This also proves the fact that suppression effects between leaf and parent categories or confusing categories are harming the performance of object detection in Open Images.

To shed more light on the effectiveness of concurrent softmax, we provide qualitative visual comparisons of different settings of utilizing concurrent softmax during both the training and testing phase in Fig. 6. As shown in the first two columns, scores of multi-labels suffer from severe inter-category inhibition, while concurrent softmax notably alleviates the suppression effects as shown in the last two columns. We also reveal the mAP improvements on single classes with co-occurrence phenomenon in Table IV.

D. Soft-Balance Sampling

Results: Table V presents the results of the proposed soft-balance sampling and other balance methods. As Open Images is a long-tailed dataset, many categories have few samples, so non-balance training only achieves 38.16 mAP. The effective number [51] is used to reweight the classification loss with the purpose of harmonizing the gradient contribution from different categories, which improves the results by 7.56 points. Other loss-based reweighting methods like [54] yield an mAP of 47.32. Class-aware sampling simply samples all categories of data uniformly at random, it remedies the data imbalance problem to a great extent and boosts the performance to 55.45. BBN also achieves a good performance of 57.83, which is much higher than loss-based reweighting methods (+10.51). We find that the balance strategy applied at the data level is more efficient than that at the loss level. We argue that the reason is the imbalance rate of Open Images (29082.8) is much higher than that in image classification datasets like iNaturalist [71] (435.1), and loss-based methods easily lead to divergence or performance degradation.

Soft-balance with the hyper-parameter λ allows us to transfer from non-balance ($\lambda = 0$) to class-aware sampling ($\lambda = 1$). Thus, we can find a point at which sufficient data from infrequent categories can be sampled to train the model without causing overfitting. Soft-balance with $\lambda = 0.7$ outperforms the class-aware sampling by 1.59 points.

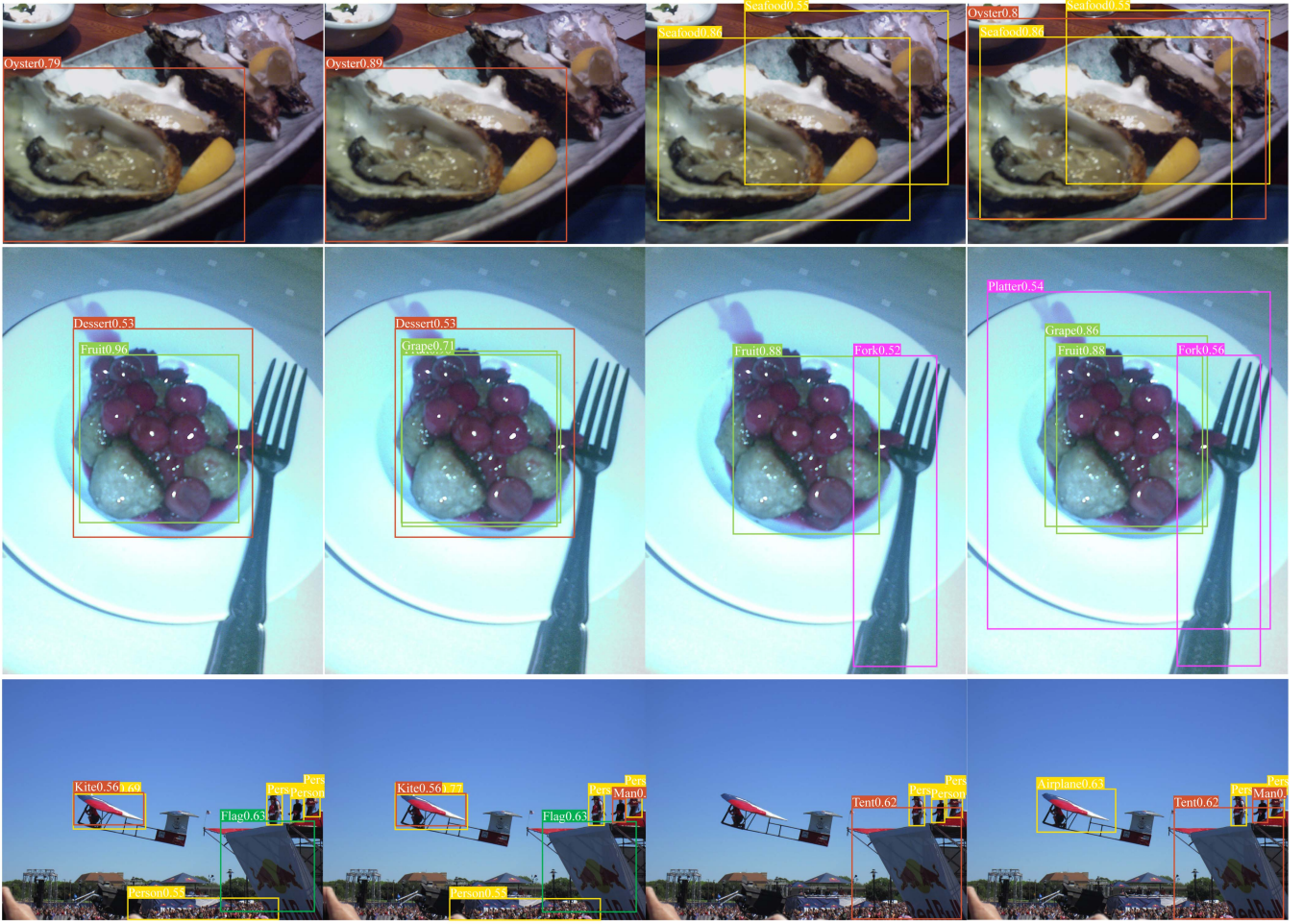


Fig. 6. Examples of detection results on the Open Images dataset. We abbreviate softmax as *sm* and concurrent softmax as *csm* here. From left to right: *sm-train+sm-test*, *sm-train+csm-test*, *csm-train+sm-test*, *csm-train+csm-test*. We set the score threshold to 0.5 for visualization.

TABLE IV
WE RANDOMLY PICK SOME CLASSES WITH CO-OCCURRENCE PHENOMENA AND SHOW THE PERFORMANCE GAIN

Source Classes	Wood-burning stove	Flute	Blender	Brown bear	Snake	Sparrow	Tea	Cello	Teddy bear	Antelope
mAP w/o CSM	0.50	0.26	0.50	0.25	0.65	0.76	0.57	0.43	0.55	0.53
mAP w/ CSM	1.00	0.63	0.83	0.50	0.89	1.00	0.80	0.65	0.75	0.72
Gain	0.50	0.36	0.33	0.25	0.24	0.24	0.23	0.22	0.20	0.18

Source Classes	Digital clock	Sword	Pineapple	Brassiere	Sea turtle	Crab	Sandal	Boot	Handbag	Sea lion	Coffee
mAP w/o CSM	0.83	0.83	0.50	0.80	0.22	0.52	0.54	0.56	0.60	0.43	0.46
mAP w/ CSM	1.00	1.00	0.67	0.97	0.38	0.67	0.68	0.70	0.73	0.46	0.58
Gain	0.17	0.17	0.17	0.17	0.16	0.15	0.14	0.14	0.13	0.13	0.12

TABLE V
THE COMPARISON OF DIFFERENT SAMPLING METHODS

Methods	λ	mAP
Non-balance	-	38.16
Effective Number [51]	-	45.72
LDAM [54]	-	47.32
Class-aware Sampling [29]	-	55.45
BNN [62]	-	57.83
Soft-balance	0.3	50.69
	0.5	56.19
	0.7	57.04
	1.0	55.45
	1.5	52.41

Models are trained in *mini-train* and evaluated on *mini-val*.

The impacts of soft-balance during training: To investigate why soft-balance is better, we show the training curves of soft-balance with different λ in Fig. 7. We can learn that the small $\lambda = 0.3$ makes it hard to achieve a good performance due to the data imbalance problem. But too large $\lambda = 1.5$ also fails to accomplish the best performance compared with the relatively smaller λ setting. Note that the mAP of $\lambda = 1.5$ is much higher than that of $\lambda = 0.5$ in the first learning rate stage (before epoch 4), but this situation reverses in subsequent train progress. This comparison proves that $\lambda = 1.5$ provides more sufficient rare categories data to train the model and achieve better performance initially. However, it runs into severe over-fitting in the

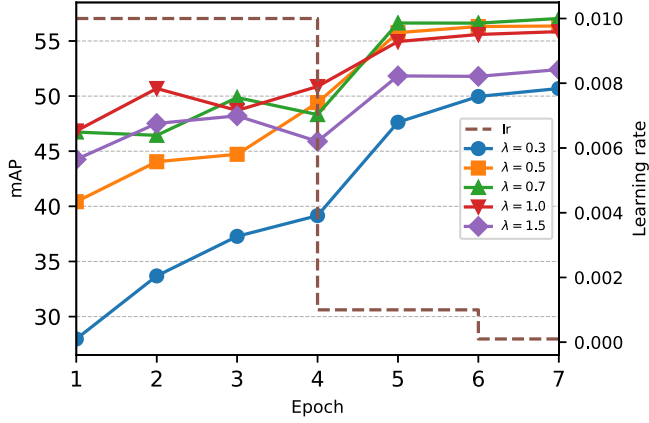


Fig. 7. Training curves of the proposed soft-balance sampling. Soft-balance with $\lambda = 0.7$ achieves the best performance.

convergence stage. The results of $\lambda = 1.0$ and $\lambda = 0.7$ further validate these rules again.

The impacts of soft-balance among categories: We further study the performance of $\lambda = 0.0$, $\lambda = 0.7$, and $\lambda = 1.0$ among categories in Fig. 8, in which the challenge validation results of 500 categories are arranged from large to few by their numbers of images. As shown in Fig. 8(a), the $\lambda = 1.0$ (orange line) outperforms the $\lambda = 0.0$ (blue line) on the later half categories which have few image samples. Although $\lambda = 1.0$ solves the data insufficiency of infrequent categories, it under-samples the frequent categories and causes the performance drop on the former half of categories. Fig. 8(b) shows that $\lambda = 0.7$ (orange line) alleviates the excessive under-sampling of the major categories compared to $\lambda = 1.0$ (blue line). On the other hand, it mitigates the over-fitting problem of infrequent categories. Therefore, the performance of $\lambda = 0.7$ is almost always better than $\lambda = 1.0$ on the full category space. We visualize some of the results to demonstrate the influence of different sampling schemes in Fig. 9.

The ablation studies on the hyperparameter λ aim to investigate the sensitivity of the proposed soft-balance sampling method to different values of λ and determine its impact on the overall performance. Meanwhile, as $\lambda = 0$ means non-balance sampling and $\lambda = 1$ means class-aware sampling, we can compare different sampling methods with our soft-balance sampling by changing the value of λ . By varying λ systematically, we aimed to analyze its effect on the model's behavior and identify an optimal value that yields the best results. Though exhaustively testing multiple choices of λ may not be feasible or practical in real-world scenarios, the purpose of the ablation studies is to explore the range of λ values and identify a value that offers a good trade-off between performance and computational cost.

E. Hybrid Training Scheduler

Table VI summarizes the results of ResNeXt152 on the Open Images Challenge dataset trained with different training schedulers. For the non-balance setting, the more epochs the model trained, the better performance the model achieves. Training a

TABLE VI
THE EFFECT OF TRAINING SCHEDULER

Method	Pretrain	Epochs	mAP
Non-balance	ImageNet	7	56.06
	ImageNet	11	59.12
	ImageNet	14	59.85
	ImageNet	16	59.95
	Scratch	20	60.70
Class-aware Sampling	ImageNet	7	64.68
	ImageNet	14	62.85
	Non-balance I14	14+7	65.60
	Non-balance S20	20+7	65.92
Soft-balance	Non-balance S20	20+7	67.09
Soft-balance*	Non-balance S20	20+7	68.23
Cosine-Annealing Sampling	Scratch	27	67.39
Cosine-Annealing Sampling*	Scratch	27	68.56

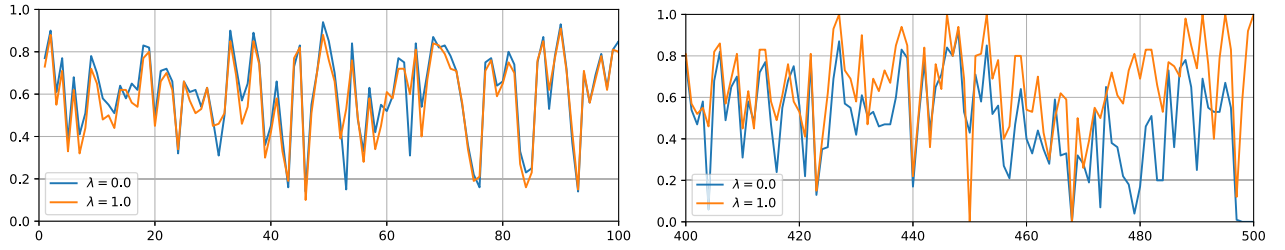
The λ of the softbalance is set to 0.7. *Non-balance I14* denotes the model of epoch 14 trained with non-balance strategy from ImageNet pretraining. *Non-balance S20* denotes the model of epoch 20 trained with the non-balance strategy from scratch. Epochs indicate the total epochs for training a model on the Open Images Challenge dataset. Since we are exploring the overall impacts of hybrid training, all models in this table are trained with the entire dataset, i.e., *full-train* and evaluated on *full-val*. Soft-balance* means that concurrent softmax is adopted in both the training and testing stages. Models are trained on *full-train* and evaluated on *full-val*.

model from scratch yields better results than finetuning from an ImageNet pre-trained model. These observations match a similar conclusion in [73]. Thus all our models are trained from scratch without ImageNet pretraining in our final training setting.

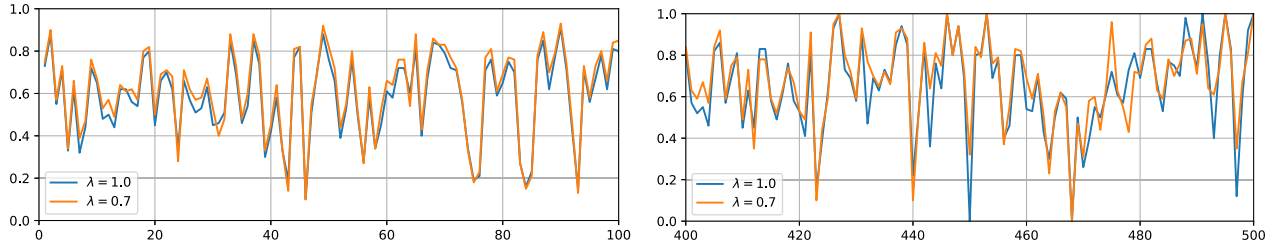
While class-aware sampling significantly boosts the performance by 8.62 points (from 56.06 to 64.68 mAP) using the ImageNet pretraining in 7 epochs setting, it suffers from an over-fitting problem, as the mAP of the model trained with 14 epochs is lower than that of 7 epochs. And frequent categories are still intensely under-sampled even applying the balance sampling. With hybrid training, class-aware sampling can achieve better performance in both non-balance ImageNet pretraining and non-balance scratch pretraining settings. Note that these improvements are not caused by more training epochs, because a longer training schedule will decrease the performance with only ImageNet pretraining. By further using a soft-balance strategy, the hybrid training with non-balance scratch is improved from 65.92 to 67.09 mAP.

As shown in the last two rows in Table VI, we simplify the two-step training procedure into a unified one with the usage of *cosine-annealing hybrid training* algorithm. This enables the models to be trained from scratch end to end and also boosts the performance for a small margin (from 67.09 to 67.39).

More importantly, cosine-annealing hybrid training performs great robustness on hyper-parameter selection. As demonstrated in Table VII, the soft-balance sampling is sensitive to the selection of λ , e.g., the mAP gap between $\lambda = 0.7$ and $\lambda = 1.1$ is 2.34. The cosine-annealing hybrid training stabilizes the choosing of λ and narrows the gap down to 0.64. We also compare our results with BBN [62] in the last row in Table VI, which also performs well. In BBN, the model is trained with a non-balanced and a re-balanced data sampler at the same time, and an α is applied for controlling the weight of each branch. To adapt BBN to our setting that trains the model from scratch, we adjust the original α scheduler $\alpha = 1 - (\frac{T_{cur}}{T_{max}})^2$ to $\lambda = 1 - \mathbb{1}(T_{cur} - T_{start})(\frac{T_{cur} - T_{start}}{T_{max} - T_{start}})^2$. Since BBN shares



(a) Non-balance (blue) versus Class-aware Sampling (orange) sorted by the number of images for the most frequent 100 categories (left) and most infrequent 100 categories (right).



(b) Class-aware Sampling (blue) versus Soft-balance with $\lambda = 0.7$ (orange) sorted by the number of images for the most frequent 100 categories (left) and most infrequent 100 categories (right).

Fig. 8. The comparison of sampling strategy among categories. (Best viewed on high-resolution display.)

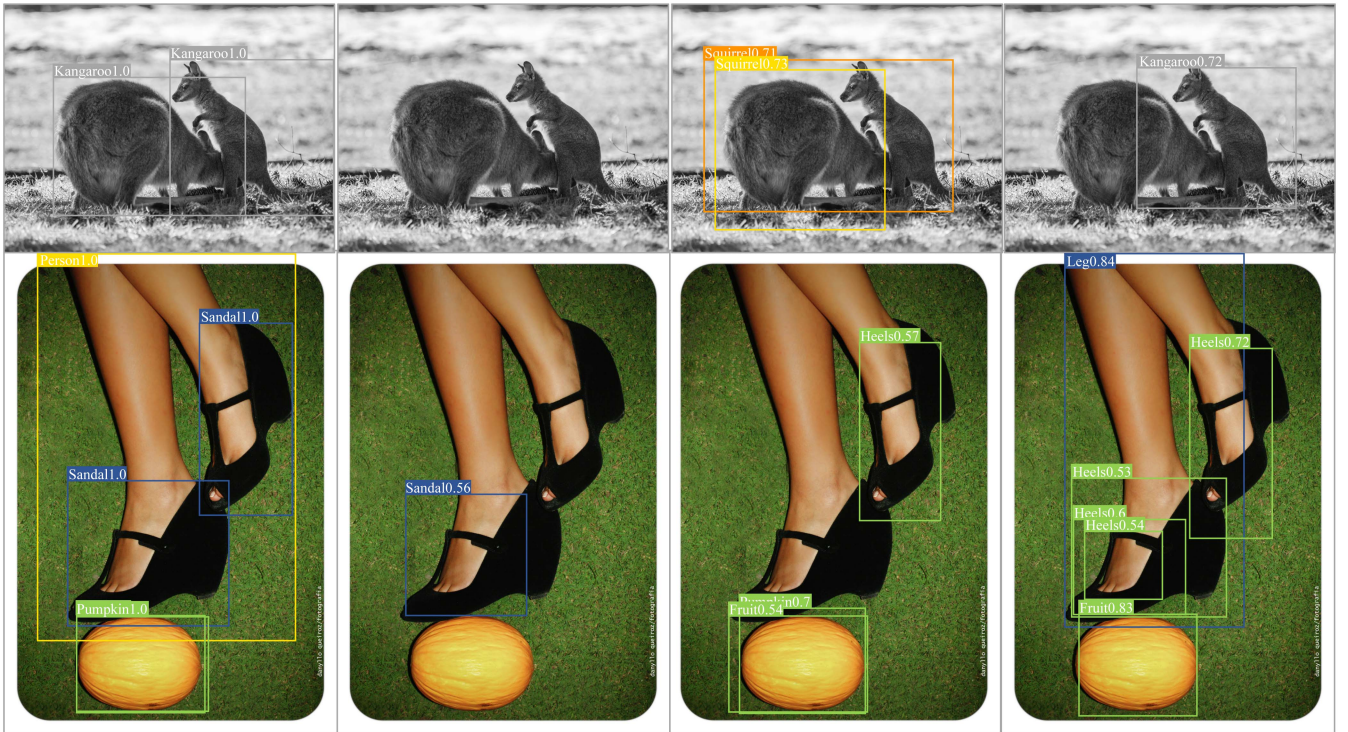


Fig. 9. Examples of detection results on the Open Images dataset. From left to right: ground truth, non-balance, class-aware balance, soft-balance. We set the score threshold to 0.5 for visualization.

the same motivation with cosine-annealing hybrid training, the core difference is how λ and α vary with epochs in respective methods, it performs well in our setting. We believe that unifying the non-balance and re-balance sampling is always a good solution for the issue of data imbalance.

Extended experiments on *minitrain* and *minival* are shown in Table VIII. The results demonstrate that when the cosine annealing sampling strategy is used, the soft-balanced sampling with $\lambda = 0.7$ will be improved by 0.15 points. The ensemble model outperforms the soft-balanced sampling with $\lambda = 0.7$ by

TABLE VII
RESULTS OF MODELS TRAINED FROM SCRATCH FOR 27 EPOCHS ON
FULL-TRAIN AND EVALUATED ON *FULL-VAL*

Method	λ	T_{start}	mAP
Default hybrid training	0.7	20	67.09
	0.9		65.73
	1.1		64.75
Cosine-Annealing hybrid training	0.7	17	67.39
	0.9		67.03
	1.1		66.75
	0.9	15	66.93
	0.9	19	67.25
BBN [62]	-	17	67.13

The detector is Faster-RCNN with x152-32×4d as the backbone, equipped with FPN.

TABLE VIII
THE RESULTS OF NON-BALANCE, SOFT-BALANCE, AND ENSEMBLE MODELS
WITH OR WITHOUT COSINE-ANNEALING SAMPLING (CAS)

Methods	λ	mAP	mAP(+CAS)
Non-balance	-	38.16	-
Soft-balance	0.3	50.69	48.79
	0.5	56.19	55.23
	0.7	57.04	57.19
	1.0	55.45	57.38
	1.5	52.41	54.67
Ensemble	-	61.25	62.17

Models are trained in *mini-train* and evaluated on *mini-val*, with ImageNet pretraining.

TABLE IX
THE SINGLE-SCALE TEST MODEL MAP ON *FULL-VAL*

Id	Architecture	Description	mAP
0	x152-32×4d+fpn+cass	$\lambda = 1.0$	65.21
1	x152-32×4d+fpn+mask	$\lambda = 1.0$	66.30
2	x152-32×4d+fpn	concurrent-softmax, $\lambda = 1.0$	68.23
3	x152-32×4d+fpn+dcnv2	$\lambda = 1.0$	66.63
4	x152-32×4d+c4	$\lambda = 1.0$	64.07
5	x152-32×4d+fpn	$\lambda = 0.4, iou$ sampling	67.65
6	x152-32×4d+fpn	$\lambda = 0.7, iou$ sampling	68.29
7	x152-32×4d+fpn	$\lambda = 1.0, iou$ sampling	65.92
8	x152-32×4d+fpn	$\lambda = 0.0, iou$ sampling	60.42
9	x152-32×4d+c4	$\lambda = 0.0, iou$ sampling	57.48
10	x152-32×16d+fpn	$\lambda = 1.5$	66.84
11	x152-32×16d+fpn	$\lambda = 1.0$	67.98
12	x152-32×16d+fpn	$\lambda = 0.7$	68.17
13	x152-32×16d+fpn+mask	$\lambda = 0.7, iou$ sampling	69.46
14	x152-32×16d+fpn	$\lambda = 0.5, iou$ sampling, obj365	69.97
15	x152-32×16d+fpn	$\lambda = 0.7, iou$ sampling, obj365	68.19
16	x152-32×16d+fpn	$\lambda = 0.0, iou$ sampling	63.24
17	x215-32×4d+fpn	$\lambda = 0.8, iou$ sampling	69.34
18	x215-32×4d+fpn	$\lambda = 0.0, iou$ sampling	65.89
19	b7+fpn	$\lambda = 1.0$	67.45
20	se154+fpn	$\lambda = 1.0$	67.16

The term *cas* denotes utilizing cascade head [19]. The term *iou-sampling* denotes the box sampling strategy noted in Libra-RCNN [72]. The term *obj365* means the model is trained with part of data from dataset object365 [69]. The term *b7* denotes our implemented efficient-b7.

TABLE X
RESULTS OF MULTI-SCALE TESTING WITH DIFFERENT HYPER-PARAMETERS
FOR BOX FILTERING

Setting	t	n	mAP
<i>model-6</i> in Table 9 with multi-scale test	0.0	100	68.77
	0.01	100	67.74
	0.0	1000	69.79
	0.01	1000	69.72

TABLE XI
RECOMMENDED VALUES OF HYPER-PARAMETERS IN OUR ENSEMBLE
STRATEGIES

Parameter	Description	Recommended Value
t_s	IoU threshold for score merging	0.42
t_b	IoU threshold for box merging	0.80
n_s	number of boxes for score merging	1
w_s	weight of score merging	1.0
w_b	weight of box merging	0.5
α	hyper-parameter for re-scoring	0.5
β	hyper-parameter for re-scoring	0.7
l	levels of hierarchical ensemble	2

TABLE XII
RESULTS OF UTILIZING VARIOUS ENSEMBLE METHODS ON VAL SET WITH 4
MODELS SELECTED FROM TABLE IX

Setting	mAP
baseline	70.45
+ re-coring	70.89
+ top- n merging	73.27
+ hierarchical ensemble	73.55

TABLE XIII
RESULTS WITH BELLS AND WHISTLES ON OPEN IMAGES PUBLIC
TEST-CHALLENGE SET

Methods	Ensemble	Public Test
2018 1st [16]		55.81
Ours		60.90
2018 1st [16]	✓	62.88
2018 2nd [29]	✓	62.16
2018 3rd	✓	61.70
Ours	✓	67.17
Baseline (ResNeXt-152)		53.88
+Class-aware Sampling		57.56
+Concurrent Softmax Loss		58.60
+Soft-balance		59.86
+Hybrid Training Scheduler		60.90
+Other Tricks		62.34
+Ensemble	✓	67.17

4.21 points. Also, when the cosine annealing sampling strategy is used, the ensemble model will be improved by 0.92 points.

F. Model Ensemble

We sustain a model pool containing 21 models to support the model ensemble in our final setting, whose mAP of *full-val* are listed in Table IX. The tricks involved in our pool includes: iou-sampling [72], dcnv2 [74], more training data from public dataset [69], more complex backbone like se-net [75] and efficient-net [76], mask-RCNN [77], cascade RCNN [19].

Model selection for model pool: In this paragraph, we explain the process of collecting models into the model pool for better ensembling. Conventionally, there are two paradigms for model selection for ensembling: models with various architecture, or models with various tricks. As in our scenario, we find both paradigms offering very limited help. For instance, *model-0*, *1*, *3*, *19*, *20* involves cascade-RCNN [19], mask-RCNN [77], deformable convolution [21], efficientNet [78] and seNet [75], but the improvement is slight and no prominent diversity is found among results of these models. Instead, we find the sampling

TABLE XIV
STATISTICS OF THE IMPLICIT CONFUSION PAIRS

Source Classes	Flashlight	Sea turtle	Bread	Hamster	Sombrero	Tiara	Antelope	Torch	Blender	Blender
Confusion Classes	Torch	Tortoise	Dessert	Mouse	Sun hat	Crown	Deer	Flashlight	Food processor	Mixer
#Source bboxes	71	1053	3691	522	618	383	1473	17	228	228
#Confusion bboxes	11	555	414	139	93	86	534	11	48	37
Source Classes	Billiard table	Tart	Fedora	Food processor	Food processor	Bookcase	Peach	Knife	Knife	
Confusion Classes	Table	Dessert	Sun hat	Blender	Mixer	Shelf	Apple	Kitchen knife	Dagger	
#Source bboxes	852	903	3447	184	184	4962	742	803	803	
#Confusion bboxes	248	157	657	48	42	930	76	161	106	
Source Classes	Ambulance	Eagle	Mixer	Mixer	Wood-burning stove	Beaker	Beehive	Cupboard		
Confusion Classes	Van	Falcon	Blender	Food processor	Fireplace	Measuring cup	Honeycomb	Cabinetry		
#Source bboxes	429	1599	203	203	289	162	492	1269		
#Confusion bboxes	171	562	37	42	67	26	147	153		
Source Classes	Grapefruit	Mechanical fan	Shotgun	Tire	Sandal	Harpichord	Violin	Leopard	Leopard	Harbor seal
Confusion Classes	Orange	Ceiling fan	Rifle	Wheel	High heels	Piano	Cello	Jaguar	Cheetah	Sea lion
#Source bboxes	1211	646	539	115719	2738	197	2613	768	768	2003
#Confusion bboxes	285	153	136	17117	351	57	275	347	370	457
Source Classes	Goose	Coffee cup	Microwave oven	Serving tray	Serving tray	Beer	Toilet paper	Kitchen knife	Lynx	
Confusion Classes	Duck	Mug	Oven	Platter	Plate	Bottle	Paper towel	Knife	Cat	
#Source bboxes	7948	5018	458	111	111	9065	356	333	222	
#Confusion bboxes	1077	1146	89	30	17	920	81	161	76	
Source Classes	Platter	Snowplow	Saucer	Insect	Ceiling fan	Jaguar	Jaguar	Taxi	Pitcher	High heels
Confusion Classes	Plate	Land vehicle	Plate	Spider	Mechanical fan	Leopard	Cheetah	Car	Jug	Sandal
#Source bboxes	3317	293	2681	8457	446	553	553	3984	329	2950
#Confusion bboxes	983	34	282	1007	153	347	218	412	121	351
Source Classes	Bust	Trombone	Bidet	Falcon	Fox	Sofa bed	Desk	Tick	Tick	Filing cabinet
Confusion Classes	Person	Trumpet	Toilet	Eagle	Dog	studio couch	Table	Beetle	Insect	Chest of drawers
#Source bboxes	986	904	411	1624	528	1417	11015	138	138	343
#Confusion bboxes	169	95	51	562	66	480	1516	14	18	69
Source Classes	Wine	Honeycomb	Doughnut	Doughnut	Chest of drawers	Dagger	Dagger	Mug	Pressure cooker	
Confusion Classes	Bottle	Beehive	Dessert	Bagel	Cabinetry	Knife	Sword	Coffee cup	Slow cooker	
#Source bboxes	14458	366	870	870	1431	350	350	2140	11	
#Confusion bboxes	1540	147	110	197	159	106	132	1146	4	
Source Classes	Tortoise	Sword	Girl	Seafood	Sea lion	Tea	Mule	Doll	Goggles	Organ
Confusion Classes	Sea turtle	Dagger	Woman	Fish	Harbor seal	Coffee	Horse	Girl	Sunglasses	Piano
#Source bboxes	1885	532	185837	2894	1747	1266	1068	6046	9087	376
#Confusion bboxes	555	132	34772	292	457	160	196	623	970	40
Source Classes	Winter melon	Winter melon	Deer	Briefcase	Briefcase	Cello	Bull	Whale	Crown	Cheetah
Confusion Classes	Watermelon	Cucumber	Antelope	Handbag	Suitcase	Violin	Cattle	Dolphin	Tiara	Leopard
#Source bboxes	40	40	3658	145	145	1894	1610	937	656	677
#Confusion bboxes	6	4	534	19	19	275	266	180	86	370
Source Classes	Cheetah	Oven	Measuring cup	Loveseat	Loveseat	Teapot	Kettle	Spider	Plate	Bagel
Confusion Classes	Jaguar	Microwave oven	Beaker	Sofa bed	Studio couch	Kettle	Teapot	Insect	Platter	Dessert
#Source bboxes	677	597	70	600	600	604	630	1876	5118	589
#Confusion bboxes	218	89	26	90	135	185	185	1007	983	100
Source Classes		Bagel	Brassiere	studio couch	Dolphin	Jug	Paper towel	Bicycle wheel		
Confusion Classes		Doughnut	Swimwear	Sofa bed	Whale	Pitcher	Toilet paper	Wheel		
#Source bboxes		589	1613	1797	1428	542	196	56138		
#Confusion bboxes		197	241	480	180	121	81	11066		

The first two rows refer to the source and its confusion category. The third row refers to the total instances of the source category, and the fourth row indicates how many instances from the confused category are labeled as sources.

strategies are more important and bring significant model diversity, especially per-categories diversity which is shown in Fig. 8. The reason we include models with and without iou-sampling into the model pool is that it brings diversity on scales, which could well compensate with other models. We also try to add a model trained with different data like *model-14,15*. Though the results of these models do not show great improvement on mAP, ensembling with them brings a great boost (+1.5% mAP).

Multi-scale test results ensemble: The setting of experiments we use in this ablation study includes Backbone of X152-32 $\times$ 4 d with FPN, soft-sampling with $\lambda = 0.7$, iou-sampling. The test scale ranges from 400 to 1800 with a step of 200. We

adjust the score threshold t and a number of kept boxes n for filtering boxes. As shown in Table X, multi-scale testing brings a marginal gain of 0.54 while $t = 0$ and $n = 100$. If we boost t from 0 to 0.01, the mAP is lowered by 1.04, which means there are some boxes among the top- n with scores less than 0.01. As we boost n from 100 to 1000, we obtain an obvious gain of 1.02 indicating that there exists a great number of true positives with very low scores. In our final setting, we use a t of 0.01 to control the total number of predicted boxes, otherwise the ensemble procedure may take too long.

Multi-model results ensemble: To clearly show the effectiveness of each design, we select four models from our model

pools in this ablation study, which are: *model-5*, 6, 7, 8 in Table IX. As shown in Table XII, the re-scoring and hierarchical ensemble schemes bring marginal performance gain while the top-*n* merging method greatly boosts the performance. All the recommended values of related hyper-parameters in our ensemble strategies are listed in Table XI.

G. Extension Results on Test-Challenge Set

In Table XIII, we validate the impact of each component of our proposed approach on the overall performance. The results show that with the proposed concurrent softmax, soft-balance and hybrid training scheduler, we achieve 67.17 mAP and 4.29 points absolute improvement compared to the first-place entry on the public test-challenge set last year. We train a ResNeXt-152 FPN with multi-scale training and testing as our baseline which achieves 53.88 mAP. After using class-aware balance, the performance is boosted to 57.56. With the help of the proposed concurrent softmax, the model achieves 58.60 mAP. The soft balance and the hybrid training scheduler lead to mAP gains of 1.26 and 1.04 points, respectively. By further using other tricks including data augmentation, loss function search, and heavier head, we achieve the best single model with a mAP of 62.34. We use ResNeXt-101, ResNeXt-152, and EfficientNet-B7 with various tricks for model ensembling. We achieve 67.17 mAP on the Open Images public test-challenge set and 4.83 absolute improvements compared to the above methods.

H. Implicit Confusion Pairs

Table XIV shows the detailed statistics of the confusion pairs explained in section 3 of the main paper. To obtain the confusion pairs, we first calculate the Intersection-over-Union(IoU) between each two instances. Then we mark those instances from different categories with an IoU higher than 0.9 as confusion bboxes. We define the confusion ratio as the number of the confusion bboxes divided by the source bboxes, and we list all 115 pairs of confusion categories whose confusion ratio is higher than 0.1.

I. Discussion

In this paper, we introduce the Concurrent Softmax method, designed to enhance object detection performance in the presence of imbalanced data distributions. The method's effectiveness is partly attributed to the concurrent rate, which leverages category-level priors to address the challenges posed by imbalanced datasets. However, it is important to note that our approach may not be ideally suited for scenarios with 'clean' data, where class distributions are relatively balanced.

Despite this limitation, the real-world applicability of our method is significant, as imbalanced data distributions are a common occurrence in practical settings. The key to successfully implementing our approach lies in the ability to conduct prior statistical analysis on new datasets. This analysis is not overly burdensome and can provide valuable insights into how

to effectively leverage statistical regularities to mitigate the issue of multi-label suppression in object detection tasks.

In summary, while our Concurrent Softmax method may have certain limitations, its potential to improve detection performance in imbalanced data scenarios is promising. Future work could explore ways to adapt the concurrent rate for more balanced datasets or to further streamline the statistical analysis process.

VI. CONCLUSION

In this paper, we investigate the multi-label problem and the imbalanced label distribution problem in large-scale object detection dataset and introduce a simple but powerful solution. We propose the concurrent softmax function to deal with explicit and implicit multi-label problems in both the training and testing stages. Our soft-balance method together with a hybrid training scheduler could effectively deal with the extremely imbalanced label distribution. We further unify the soft-balance sampling and hybrid training scheduler into one form. Finally, we introduce our well-designed model ensemble strategies. We sincerely wish more promising studies spring up on large-scale object detection.

REFERENCES

- [1] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2009, pp. 248–255.
- [2] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2012, pp. 1106–1114.
- [3] D. Mahajan et al., "Exploring the limits of weakly supervised pretraining," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 185–201.
- [4] Q. Xie, M.-T. Luong, E. Hovy, and Q. V. Le, "Self-training with noisy student improves imagenet classification," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 10687–10698.
- [5] M. Everingham, L. Van Gool, C. K. Williams, J. Winn, and A. Zisserman, "The Pascal visual object classes (VOC) challenge," *Int. J. Comput. Vis.*, vol. 88, no. 2, pp. 303–338, 2010.
- [6] T.-Y. Lin et al., "Microsoft COCO: Common objects in context," in *Proc. Eur. Conf. Comput. Vis.*, Springer, 2014.
- [7] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2015, pp. 91–99.
- [8] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 779–788.
- [9] W. Liu et al., "SSD: Single shot multibox detector," in *Proc. Eur. Conf. Comput. Vis.*, 2016, pp. 21–37.
- [10] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2017, pp. 2999–3007.
- [11] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis.*, Springer, 2020, pp. 213–229.
- [12] X. Zhu, W. Su, L. Lu, B. Li, X. Wang, and J. Dai, "Deformable detr: Deformable transformers for end-to-end object detection," in *Proc. Int. Conf. Learn. Representation*, 2021, pp. 1–11.
- [13] G. Zhang, Z. Luo, K. Cui, S. Lu, and E. P. Xing, "Meta-DETR: Image-level few-shot detection with inter-class correlation exploitation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 11, pp. 12832–12843, Nov. 2023.
- [14] A. Kuznetsova et al., "The open images dataset V4: Unified image classification, object detection, and visual relationship detection at scale," *Int. J. Comput. Vis.*, vol. 128, no. 7, pp. 1956–1981, 2020.

- [15] J. Peng, X. Bu, M. Sun, Z. Zhang, T. Tan, and J. Yan, "Large-scale object detection in the wild from imbalanced multi-labels," in *IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 9706–9715.
- [16] T. Akiba, T. Kerola, Y. Niitani, T. Ogawa, S. Sano, and S. Suzuki, "Pfdet: 2nd place solution to open images challenge 2018 object detection track," 2018, *arXiv: 1809.00778*.
- [17] H. Law and J. Deng, "CornerNet: Detecting objects as paired keypoints," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 734–750.
- [18] X. Zhou, D. Wang, and P. Krähenbühl, "Objects as points," 2019, *arXiv: 1904.07850*.
- [19] Z. Cai and N. Vasconcelos, "Cascade R-CNN: Delving into high quality object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 6154–6162.
- [20] X. Lu, B. Li, Y. Yue, Q. Li, and J. Yan, "Grid R-CNN," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 7363–7372.
- [21] J. Dai et al., "Deformable convolutional networks," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2017, pp. 764–773.
- [22] Y. Li, Y. Chen, N. Wang, and Z. Zhang, "Scale-aware trident networks for object detection," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2019, pp. 6054–6063.
- [23] J. Peng, M. Sun, Z. Zhang, T. Tan, and J. Yan, "Pod: Practical object detection with scale-sensitive network," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2019, pp. 9607–9616.
- [24] Z. Ge, S. Liu, F. Wang, Z. Li, and J. Sun, "YOLOX: Exceeding YOLO series in 2021," 2021, *arXiv:2107.08430*.
- [25] S. Chen, P. Sun, Y. Song, and P. Luo, "DiffusionDet: Diffusion model for object detection," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2023, pp. 19830–19843.
- [26] Z. Zhang, C. Pan, and J. Peng, "Delving into the effectiveness of receptive fields: Learning scale-transferrable architectures for practical object detection," *Int. J. Comput. Vis.*, vol. 130, no. 4, pp. 970–989, 2022.
- [27] Y. Niitani, T. Akiba, T. Kerola, T. Ogawa, S. Sano, and S. Suzuki, "Sampling techniques for large-scale object detection from sparsely annotated objects," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 6510–6518.
- [28] Z. Wu, N. Bodla, B. Singh, M. Najibi, R. Chellappa, and L. S. Davis, "Soft sampling for robust object detection," 2018, *arXiv: 1806.06986*.
- [29] Y. Gao et al., "Solution for large-scale hierarchical object detection datasets with incomplete annotation and data imbalance," 2018, *arXiv: 1810.06208*.
- [30] Y. Gong, Y. Jia, T. Leung, A. Toshev, and S. Ioffe, "Deep convolutional ranking for multilabel image annotation," 2013, *arXiv:1312.4894*.
- [31] J. Wang, Y. Yang, J. Mao, Z. Huang, C. Huang, and W. Xu, "CNN-RNN: A unified framework for multi-label image classification," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 2285–2294.
- [32] J. Zhang, Q. Wu, C. Shen, J. Zhang, and J. Lu, "Multilabel image classification with regional latent semantic dependencies," *IEEE Trans. Multimedia*, vol. 20, no. 10, pp. 2801–2813, Oct. 2018.
- [33] X. Li, F. Zhao, and Y. Guo, "Multi-label image classification with a probabilistic label enhancement model," in *Proc. 13th Conf. Uncertainty Artif. Intell.*, 2014, pp. 430–439.
- [34] M. Tan et al., "Learning graph structure for multi-label image classification via clique generation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2015, pp. 4100–4109.
- [35] J. Wu, A. Guo, V. S. Sheng, P. Zhao, Z. Cui, and H. Li, "Adaptive low-rank multi-label active learning for image classification," in *Proc. ACM Int. Conf. Multimedia*, 2017, pp. 1336–1344.
- [36] Z.-M. Chen, X.-S. Wei, P. Wang, and Y. Guo, "Multi-label image recognition with graph convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 5177–5186.
- [37] V. O. Yazici, A. Gonzalez-Garcia, A. Ramisa, B. Twardowski, and J. v. d. Weijer, "Orderless recurrent models for multi-label classification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 13440–13449.
- [38] X. Cheng et al., "MLTR: Multi-label classification with transformer," in *Proc. IEEE Int. Conf. Multimedia Expo*, 2022, pp. 1–6.
- [39] M. R. Boutell, J. Luo, X. Shen, and C. M. Brown, "Learning multi-label scene classification," *Pattern Recognit.*, vol. 37, no. 9, pp. 1757–1771, 2004.
- [40] H. Hu, G.-T. Zhou, Z. Deng, Z. Liao, and G. Mori, "Learning structured inference neural networks with label relations," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 2960–2968.
- [41] Q. Li, M. Qiao, W. Bian, and D. Tao, "Conditional graphical lasso for multi-label image classification," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 2977–2986.
- [42] Z. Wang, T. Chen, G. Li, R. Xu, and L. Lin, "Multi-label image recognition by recurrently discovering attentional regions," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2017, pp. 464–472.
- [43] Y. Wang et al., "Multi-label classification with label graph superimposing," in *Proc. AAAI Conf. Artif. Intell.*, 2020, pp. 12265–12272.
- [44] S.-F. Chen, Y.-C. Chen, C.-K. Yeh, and Y.-C. Wang, "Order-free RNN with visual attention for multi-label classification," in *Proc. AAAI Conf. Artif. Intell.*, 2018, pp. 6714–6721.
- [45] J. Lanchantin, T. Wang, V. Ordonez, and Y. Qi, "General multi-label image classification with transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 16478–16488.
- [46] Z.-M. Chen, Q. Cui, B. Zhao, R. Song, X. Zhang, and O. Yoshie, "SST: Spatial and semantic transformers for multi-label image recognition," *IEEE Trans. Image Process.*, vol. 31, pp. 2570–2583, 2022.
- [47] L. Shen, Z. Lin, and Q. Huang, "Relay backpropagation for effective learning of deep convolutional neural networks," in *Proc. Eur. Conf. Comput. Vis.*, 2016, pp. 467–482.
- [48] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "Smote: Synthetic minority over-sampling technique," *J. Artif. Intell. Res.*, vol. 16, pp. 321–357, 2002.
- [49] W. Ouyang, X. Wang, C. Zhang, and X. Yang, "Factors in finetuning deep model for object detection with long-tail distribution," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 864–873.
- [50] Y. Zou, Z. Yu, B. Vijaya Kumar, and J. Wang, "Unsupervised domain adaptation for semantic segmentation via class-balanced self-training," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 297–313.
- [51] Y. Cui, M. Jia, T.-Y. Lin, Y. Song, and S. Belongie, "Class-balanced loss based on effective number of samples," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 9268–9277.
- [52] C. Huang, Y. Li, C. Change Loy, and X. Tang, "Learning deep representation for imbalanced classification," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 5375–5384.
- [53] Y.-X. Wang, D. Ramanan, and M. Hebert, "Learning to model the tail," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 7029–7039.
- [54] K. Cao, C. Wei, A. Gaidon, N. Aréchiga, and T. Ma, "Learning imbalanced datasets with label-distribution-aware margin loss," in *Proc. Adv. Neural Inf. Process. Syst.*, 2019, pp. 1565–1576.
- [55] S. Wang, W. Liu, J. Wu, L. Cao, Q. Meng, and P. J. Kennedy, "Training deep neural networks on imbalanced data sets," in *Proc. Int. Joint Conf. Neural Netw.*, 2016, pp. 4368–4374.
- [56] Q. Dong, S. Gong, and X. Zhu, "Imbalanced deep learning by minority class incremental rectification," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 41, no. 6, pp. 1367–1381, Jun. 2019.
- [57] H.-P. Chou, S.-C. Chang, J.-Y. Pan, W. Wei, and D.-C. Juan, "Remix: Rebalanced mixup," in *Proc. Eur. Conf. Comput. Vis. Workshops*, Glasgow, U.K., Springer, 2020, pp. 95–110.
- [58] P. Chu, X. Bian, S. Liu, and H. Ling, "Feature space augmentation for long-tailed data," in *Proc. Eur. Conf. Comput. Vis.*, Glasgow, U.K., Springer, 2020, pp. 694–710.
- [59] B. Liu, H. Li, H. Kang, G. Hua, and N. Vasconcelos, "Breadcrumbs: Adversarial class-balanced sampling for long-tailed recognition," in *Proc. Eur. Conf. Comput. Vis.*, Springer, 2022, pp. 637–653.
- [60] Z. Zhong, J. Cui, S. Liu, and J. Jia, "Improving calibration for long-tailed recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 16489–16498.
- [61] Y. Zang, C. Huang, and C. C. Loy, "FASA: Feature augmentation and sampling adaptation for long-tailed instance segmentation," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2021, pp. 3457–3466.
- [62] B. Zhou, Q. Cui, X. Wei, and Z. Chen, "BBN: Bilateral-branch network with cumulative learning for long-tailed visual recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 9716–9725.
- [63] A. Shrivastava, A. Gupta, and R. Girshick, "Training region-based object detectors with online hard example mining," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 761–769.
- [64] K. Cao, C. Wei, A. Gaidon, N. Aréchiga, and T. Ma, "Learning imbalanced datasets with label-distribution-aware margin loss," in *Proc. Adv. Neural Inf. Process. Syst.*, 2019, pp. 1565–1576.
- [65] B. Li et al., "Equalized focal loss for dense long-tailed object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 6980–6989.
- [66] M. Buda, A. Maki, and M. A. Mazurowski, "A systematic study of the class imbalance problem in convolutional neural networks," *Neural Netw.*, vol. 106, pp. 249–259, 2018.
- [67] J. M. Johnson and T. M. Khoshgoftaar, "Survey on deep learning with class imbalance," *J. Big Data*, vol. 6, no. 1, pp. 1–54, 2019.

- [68] Y. Zhang, B. Kang, B. Hooi, S. Yan, and J. Feng, "Deep long-tailed learning: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 9, pp. 10795–10816, Sep. 2023.
- [69] S. Shao et al., "Objects365: A large-scale, high-quality dataset for object detection," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2019, pp. 8429–8438.
- [70] R. Girshick, I. Radosavovic, G. Gkioxari, P. Dollár, and K. He, "Detectron," 2018. [Online]. Available: <https://github.com/facebookresearch/detectron>
- [71] G. Van Horn et al., "The inaturalist species classification and detection dataset," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 8769–8778.
- [72] J. Pang, K. Chen, J. Shi, H. Feng, W. Ouyang, and D. Lin, "Libra R-CNN: Towards balanced learning for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 821–830.
- [73] K. He, R. Girshick, and P. Dollár, "Rethinking Imagenet pre-training," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2019, pp. 4918–4927.
- [74] X. Zhu, H. Hu, S. Lin, and J. Dai, "Deformable convnets V2: More deformable, better results," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 9308–9316.
- [75] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 7132–7141.
- [76] M. Tan and Q. V. Le, "Efficientnet: Rethinking model scaling for convolutional neural networks," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 6105–6114.
- [77] K. He, G. Gkioxari, P. Dollar, and R. Girshick, "Mask R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2017, pp. 2980–2988.
- [78] M. Tan and Q. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 6105–6114.



Cong Pan received the BEng degree from the Nanjing University of Aeronautics and Astronautics (NUAA), in 2018, and the PhD degree from the Institute of Automation, Chinese Academy of Sciences (CASIA), in 2024. Her current research interests include object detection, autonomous driving, computer vision, and pattern recognition.



Junran Peng received the BS degree from Tsinghua University and the PhD degree from the Chinese Academy of Science. He has published many papers in the international journal and conferences including *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *International Journal of Computer Vision*, *ACL*, *ICCV*, *NeurIPS*, *CVPR*, and etc. His research interests include computer vision, embodied AI, AutoML, MLOps, and AI system.



Xingyuan Bu received the BS and MS degrees in computer science from the Beijing Institute of Technology (BIT), Beijing, China, in 2015, and 2018. Currently he is a researcher with Alibaba. His research interests include computer vision and machine learning.



Zhaoxiang Zhang (Senior Member, IEEE) received the bachelor's degree in circuits and systems from the University of Science and Technology of China (USTC), in 2004, and the PhD degree from the National Laboratory of Pattern Recognition (NLPR), Institute of Automation, Chinese Academy of Sciences, in 2009, under the supervision of professor Tieniu Tan. He is currently a professor with the Center for Research on Intelligent Perception and Computing, Institute of Automation, Chinese Academy of Sciences (CASIA), Beijing, China. In 2009, he joined

the School of Computer Science and Engineering, Beihang University, as an assistant professor (2009 to 2011), an associate professor (2012 to 2015) and the vice-director of the Department of Computer application technology (2014 to 2015). In 2015, he returned to the Institute of Automation, Chinese Academy of Sciences to be a professor. His major research interests include pattern recognition, computer vision, machine learning and bio-inspired visual computing. He has published more than 150 papers in reputable conferences and journals. He has won the best paper awards in several conferences and championships in international competitions. He has served as the area chair, senior PC of many international conferences like *CVPR*, *ICCV*, *AAAI*, and *IJCAI*. He is now serving as the associate editor of *IEEE Transactions on Circuits and Systems for Video Technology*, *Pattern Recognition* and *Neurocomputing*.